**13310 Spring 2025: Problem Set 1** Agustin Esteva
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# Problem 1

**Payoff Matrix:**

| 1/2 | Work | Shirk |
|---|---|---|
| Work | $r-c, \theta-c$ | $-c, 0$ |
| Shirk | $0, -c$ | $0, 0$ |

Player 1's value of collaboration $r \in (0,1)$ is common knowledge, while player 2's value $\theta \in [0,1]$ is drawn from the uniform distribution and observed only by player 2.

(a) Is there any Bayesian Nash Equilibrium (BNE) in which player 1 always chooses to shirk? If there is, what is player 2's strategy in this BNE?

SOLUTION: Consider the **strategy in which player 1 chooses to shirk always** and **player 2 chooses to shirk regardless of type** $\theta$. Denote by $a_i$ the action taken by player $i$. Note that $a_i \in \{W, S\}$.

Fix player 2 with this strategy. Then player 1 receives a payoff from shirking of

$$U_1(a_1 = S|\theta) = U_1(a_1 = S) = 0.$$

Player 1 receives a payoff from working of

$$U_1(a_1 = W \mid \theta) = U_1(a_1 = W) = -c.$$

This will be optimal for player 1 as long as $\boxed{0 \geq -c}$, which is always true.

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(b) Assume there is a BNE in which player 1 always chooses to work. What is player 2's strategy in this BNE?

SOLUTION: Fix player 1 with this strategy. Then player 2 receives a payoff from shirking of

$$U_2(a_2 = S \mid \theta) = U_2(a_2 = S) = 0.$$

Player 2 receives a payoff from working of

$$U_2(a_2 = W \mid \theta) = -c.$$

This will be optimal for player 2 as long as $0 \geq -c$, which is always true.

Thus, this strategy denotes a BNE. The strategy is for player 1 to always work and for player 2 to work if, and only if, $\theta \geq c$.

To see this, first fix player 1, then player 2's payoff matrix is given by

| | Work | Shirk |
|---|---|---|
| Work | $\theta - c$ | 0 |

the expected payoffs are given by

$$\mathbb{E}[W_2] = \theta - c, \quad \mathbb{E}[S_2] = 0,$$

and so player 2 will work if, and only if, $\theta - c \geq 0 \implies \boxed{\theta \geq c.}$ That is, if the payoff of player 2 is greater than the cost, then he will always work.

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(c) Derive the condition in terms of $c$ and $r$ under which there exists a BNE in which player 1 always chooses to work.

SOLUTION: Fix player 1 2 with the strategies as above. Consider that since $\theta \sim U([0,1])$, then if $c \in [0,1]$

$$\mathbb{P}\{\theta < c\} = c, \quad \mathbb{P}\{\theta \geq c\} = 1 - c.$$

Then the payoff player 1 receives from working is

$$\mathbb{E}[W] = (r-c)\mathbb{P}\{\theta \geq c\} + (-c)\mathbb{P}\{\theta < c\} = (r-c)(1-c) - c^2 = r - rc - c + c^2 - c^2 = r(1-c) - c.$$

The payoff player 1 receives from shirking is

$$\mathbb{E}[S_1] = 0.$$

Thus, we need $r(1-c) \geq c$ and so player 1 will work if $\boxed{r \geq \dfrac{c}{1-c}}$, which is always met, but by our strategy profile, player 2 always works.

Thus, this strategy denotes a BNE. Consider when $r \geq \frac{c}{1-c}$. Then the strategy profile where player 1 always works and player 2 works if, and only if, $\boxed{\theta \geq c}$ constitutes a BNE. ■

(d) Assume there is a BNE in which player 1 chooses to work with probability $p \in (0,1)$. What is player 2's strategy in this BNE?

SOLUTION: We claim that the strategy with which player 1 chooses to work with probability $p$ and shirks with probability $1-p$ and player 2 works if and only if $\theta \geq \frac{c}{p}$ constitutes a BNE.

First, fix player 1s strategy. Then the payoffs of player 2 are given by

$$\mathbb{E}[W_2] = (\theta - c)p + (-c)(1-p) = \theta p - cp - c + cp = \theta p - c$$

$$\mathbb{E}[S_2] = 0.$$

Thus, player 2 will work if, and only if,

$$\theta p - c \geq 0 \implies \boxed{\theta \geq \frac{c}{p}}.$$

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(e) Derive the condition in terms of $c$ and $r$ under which there exists a BNE in which player 1 chooses to work with some probability $p \in (0,1)$, and calculate $p$ as a function of $c$ and $r$.

(Hint: $p \in (0,1)$ requires player 1 to be indifferent between working and shirking.)

SOLUTION: Fix player 2s strategy from above to work iff $\theta \geq \frac{c}{p}$ Then the payoffs of player 1 are given by

$$\mathbb{E}[W_1] = (r-c)\mathbb{P}\{\theta \geq \frac{c}{p}\} + (-c)\mathbb{P}\{\theta < \frac{c}{p}\} = (r-c)(1-\frac{c}{p}) - \frac{c^2}{p} = r - \frac{rc}{p} - c$$

$$\mathbb{E}[S_1] = 0,$$

and so player 1 will work if, and only if,

$$r - \frac{rc}{p} - c \geq 0 \iff rp - rc - cp \geq 0 \iff p(r-c) \geq rc \iff p \geq \frac{rc}{r-c}$$

Thus,

$$\boxed{p = \frac{rc}{r-c}}$$

is where player 1 is indifferent between working and shirking. Since $p$ is a probability, we have that

$$0 < p < 1 \implies 0 < \frac{rc}{r-c} < 1 \implies rc < r - c \implies \boxed{\frac{c}{1-c} < r}$$

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## Problem 2

Consider an information structure with a prior $\mu \in \Delta(\Theta)$ and a type generating function $\pi : \Theta \to \Delta(\Theta)$. Prove that if this information structure is conditionally independent, then

$$\pi(\theta, s_{-i} \mid s_i) = \pi(\theta \mid s_i)\pi(s_{-i} \mid \theta) = \frac{\pi(s_{-i} \mid \theta)\pi(s_i \mid \theta)\mu(\theta)}{\sum_{\theta\in\Theta} \pi(s_i \mid \theta)\mu(\theta)}$$

SOLUTION: For the second equality, consider that by Bayes' rule

$$\pi(\theta \mid s_i) = \frac{\pi(\theta, s_i)}{\pi(s_i)}.$$

For the numerator, we have that again, by Bayes' rule:

$$\pi(\theta, s_i) = \pi(s_i \mid \theta)\pi(\theta) = \pi(s_i \mid \theta)\mu(\theta).$$

For the denominator, we use the law of total probability,

$$\pi(s_i) = \sum_{\theta\in\Theta} \pi(s_i \mid \theta)\pi(\theta) = \sum_{\theta\in\Theta} \pi(s_i \mid \theta)\mu(\theta).$$

Putting it all together, we see that

$$\pi(\theta \mid s_i) = \frac{\pi(s_i \mid \theta)\mu(\theta)}{\sum_{\theta\in\Theta} \pi(s_i \mid \theta)\mu(\theta)}.$$

For the first equality, we again use Bayes' rule:

$$\begin{aligned}
\pi(\theta, s_{-i} \mid s_i) &= \frac{\pi(\theta, s_{-i}, s_i)}{\pi(s_i)} \\
&= \frac{\pi(s_i)\frac{\pi(\theta,s_i)}{\pi(s_i)}\frac{\pi(\theta,s_{-i},s_i)}{\pi(\theta,s_i)}}{\pi(s_i)} \\
&= \pi(\theta \mid s_i)\pi(s_{-i} \mid \theta, s_i) \\
&= \pi(\theta \mid s_i)\frac{\pi(s_{-i}, \theta, s_i)}{\pi(\theta, s_i)} \\
&= \pi(\theta \mid s_i)\frac{\pi(s_{-i}s_i \mid \theta)\pi(\theta)}{\pi(s_i \mid \theta)\pi(\theta)} \\
&= \pi(\theta \mid s_i)\frac{\pi(s_{-i} \mid \theta)\pi(s_i \mid \theta)\pi(\theta)}{\pi(s_i \mid \theta)\pi(\theta)} \\
&= \pi(\theta \mid s_i)\pi(s_{-i} \mid \theta).
\end{aligned}$$

Note that we used conditional indpendence for the second to last equality. ■

## Problem 3

Two armies, A and B, are about to fight against each other, each consists of a continuum of soldiers. Soldier in army A is indexed by $i \in [0,1]$ while soldier in army B is indexed by $j \in [0,1]$. Each soldier $i$ chooses whether to fight for group A or to desert, $a_i = 0, 1$, where $a_i = 1$ indicates fighting for group A. Each soldier $j$ chooses whether to fight for group B or to desert, $b_j = 0, 1$, where $b_j = 1$ indicates fighting for group B. Fighting turnout in army A and B are respectively

$$A = \int_0^1 a_i \, di, \quad B = \int_0^1 b_j \, dj$$

A soldier's payoff depends on which army wins and on whether he/she fights for the army. Specifically, each $i$ gets

| | A wins | B wins |
|---|---|---|
| $a_i = 1$ | $r_A - c_A$ | $-c_A$ |
| $a_i = 0$ | $0$ | $0$ |

where $c_A > 0$ is the cost of fighting for army A and $r_A > c_A$ is the reward. Note that a soldier in group A gets the reward iff he participates the fight and army A defeats army B. Similarly, each $j$ gets

| | A wins | B wins |
|---|---|---|
| $b_j = 1$ | $-c_B$ | $r_B - c_B$ |
| $b_j = 0$ | $0$ | $0$ |

where $c_B > 0$ and $r_B > c_B$.

A defeats B iff

$$A - B \geq \theta,$$

where $\theta \in \mathbb{R}$ measures the advantage of B against A due to factors other than soldier turnout.

Nobody knows $\theta$ and the prior is uniform. Each $i$ receives a signal

$$s_i = \theta + \epsilon_i$$

and each $j$ receives a signal

$$s_j = \theta + \epsilon_j,$$

where $\epsilon_i$ and $\epsilon_j$ are i.i.d. and drawn from a c.d.f. $F$.

(a) (1 point) If $\theta > 1$, who would win and what would each soldier do? If $\theta < -1$, who would win and what would each soldier do?

SOLUTION: Suppose $\theta > 1$, then army $A$ will defeat $B$ iff $A - B \geq \theta$. As defined, $\max A = 1$ and $\min B = 0$, that is, the maximum $\max A - B = 1$, and so $A - B < \theta$ for any possible combination of the armies. Thus, **army $A$ will lose no matter what.**

Given this, then for any $i \in [0, 1]$, if $i$ fights, he receives a payoff of $-c_A$ with probability 1. If he surrenders, then he receives a payoff 0 with probability 1. Since $0 \geq -c_A$, **then $i$ will surrender no matter the size of $B$.**

For any $j \in [0, 1]$, if $j$ fights, he receives a payoff of $r_B - c_B$ with probability 1. If he surrenders, then he receives a payoff of 0. Since $r_B - c_B > 0$ by assumption, **then $j$ will fight no matter what.**

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Suppose $\theta < -1$. Then since $\min A = 0$ and $\max B = 1$, we have that $\min A - B = -1$. Thus, $A - B \geq -1 > \theta$ no matter the combination of the armies, and so $A$ **will win no matter what.** Using symmetrical logic as the above case, **a soldier $i$ will fight unconditionally** and **a soldier $j$ will surrender unconditionally.** ■

(b) (1 point) What is a critical state of this game?

SOLUTION: We may guess that army $A$ will win iff $\boxed{\theta \leq \theta^*}$ for some $\theta^* \leq 1$. This is because the lower $\theta$ is, the lower the advantage of $B$ against $A$, and so $A - B$ has a higher chance of being above $\theta$. ■

(c) (2 points) Suppose we have a critical state $\theta^*$. What would soldier $i$ in army $A$ do after observing $s_i$? What would soldier $j$ in army $B$ do after observing $s_j$?

SOLUTION: Fix army $B$ and consider soldier $i$. He receives payoff of $r_A - c_A$ when they win, and $-c_A$ when $A$ loses. The probability of $A$ winning (given the information that $i$ has) is (denoting a win by W and a loss by L),

$$\mathbb{P}\{W \mid s_i\} = \mathbb{P}\{\theta \leq \theta^* \mid s_i\} = \mathbb{P}\{s_i - \varepsilon_i \leq \theta^*\} = \mathbb{P}\{\varepsilon_i > s_i - \theta^*\} = 1 - F(s_i - \theta^*) = 1 - \mathbb{P}\{L \mid s_i\}.$$

Thus, the payoff of $i$ going to war is

$$\mathbb{E}[a_i = 1 \mid s_i] = (r_A - c_A)(1 - F(s_i - \theta^*)) + (-c_A)(F(s_i - \theta^*) = r_A(1 - F(s_i - \theta^*) - c_A.$$

The payoff of $i$ surrendering is

$$\mathbb{E}[a_i = 0 \mid s_i] = 0.$$

Thus, player $i$ fights in this war if and only if

$$\begin{aligned} & r_A(1 - F(s_i - \theta^*) - c_A \geq 0 \\ \implies & F(s_i - \theta^*) \leq 1 - \frac{c_A}{r_A} \\ \implies & s_i - \theta^* \leq F^{-1}(1 - \frac{c_A}{r_A}) \end{aligned}$$

$$\implies \boxed{s_i \leq \theta^* + F^{-1}(1 - \frac{c_A}{r_A})} =: k_A.$$

Thus, player $i$ fights iff $s_i \leq \theta^* + F^{-1}(1 - \frac{c_A}{r_A}) = k_A$. Using symmetrical logic, player $j$ will fight iff

$$\boxed{s_j \geq \theta^* + F^{-1}(\frac{c_B}{r_B})} =: k_B.$$

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(d) (2 points) Prove that given the strategies of $i$'s and $j$'s you derive in 2.3, there must be a critical state. (Hint: use the strong law of large numbers).

SOLUTION: We assume that each soldier makes their decision independently. The strong law of large numbers then states that

$$\int_0^1 a_i di = \mathbb{E}[a_i \mid \theta]$$

Thus, using the definition of expectation,

$$\begin{aligned} A &= \int_0^a a_i di \\ &= \mathbb{E}[a_i \mid \theta] \\ &= \mathbb{P}\{a_i = 1 \mid \theta\} \\ &= \mathbb{P}\{s_i \leq k_A \mid \theta\} \\ &= \mathbb{P}\{\theta + \varepsilon_i \leq k_A\} \\ &= F(k_A - \theta) \end{aligned}$$

Similarly,

$$\begin{aligned} B &= \int_0^1 b_j dj \\ &= \mathbb{E}[b_i \mid \theta] \\ &= \mathbb{P}\{b_i = 1 \mid \theta\} \\ &= \mathbb{P}\{s_j \geq k_B \mid \theta\} \\ &= \mathbb{P}\{\theta + \epsilon_j \geq k_B\} \end{aligned}$$

Symmetrically, we have that

$$B = 1 - F(k_B - \theta).$$

Then army $A$ will win and $B$ will lose when

$$A - B \geq \theta \iff F(k_A - \theta) - (1 - F(k_B - \theta)) \geq \theta \iff F(k_A - \theta) + F(k_B - \theta) \geq 1 + \theta.$$

We claim there exists some $\theta^*$ such that

$$F(k_A - \theta^*) + F(k_B - \theta^*) = 1 + \theta^*.$$

To see this, define an auxiliary function $\varphi : \Theta \to \mathbb{R}$ by

$$\varphi(\theta) := F(k_A - \theta) + F(k_B - \theta) - 1 - \theta.$$

It suffices to show that there is some $\theta^*$ such that $\varphi(\theta^*) = 0$. Recall a few facts about any cdf:

- $F$ is monotonically increasing;
- $F$ is right continuous, but assume that it is continuous since we are dealing with continuous random variables;
- $\lim_{x\to-\infty} F(x) = 0$, and $\lim_{x\to\infty} F(x) = 1$.

Thus, we have that $\varphi$ is continuous over $\Theta$ with

$$\lim_{\theta\to-\infty} \varphi(\theta) = \lim_{\theta\to-\infty} F(k_A - \theta) + \lim_{\theta\to-\infty} F(k_B - \theta) - 1 - \lim_{\theta\to-\infty} \theta = 1 + 1 - 1 + \infty = \infty.$$

Similarly,

$$\lim_{\theta\to\infty} \varphi(\theta) = -\infty$$

By the IVT, there is some $\theta^* \in \Theta$ such that $\varphi(\theta^*) = 0$. That is,

$$F(k_A - \theta^*) + F(k_B - \theta^*) = 1 + \theta^*.$$

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(e) (2 points) Solve the critical state in BNE and discuss how it relates to $r_A, c_A$ and $r_B, c_B$. Comment on the solution as if you were commanding an army.

SOLUTION: We have 3 equations, 3 unknowns:

$$F(k_A - \theta^*) + F(k_B - \theta^*) = 1 + \theta^* \tag{1}$$

$$\theta^* + F^{-1}(1 - \frac{c_A}{r_A}) = k_A \tag{2}$$

$$\theta^* + F^{-1}(\frac{c_B}{r_B}) = k_B \tag{3}$$

Plug in (2) and (3) into (1):

$$\begin{aligned}
F(k_A - \theta^*) + F(k_B - \theta^*) &= F(\theta^* + F^{-1}(1 - \frac{c_A}{r_A}) - \theta^*) + F(\theta^* + F^{-1}(\frac{c_B}{r_B}) - \theta^*) \\
&= F(F^{-1}(1 - \frac{c_A}{r_A})) + F(F^{-1}(\frac{c_B}{r_B})) \\
&= 1 - \frac{c_A}{r_A} + \frac{c_B}{r_B} \\
&= 1 + \theta^*
\end{aligned}$$

Thus,

$$\boxed{\theta^* = \frac{c_B}{r_B} - \frac{c_A}{r_A}}$$

Suppose I am commander A. I would want $\theta^*$ to be as high as possible, so then I would want $\frac{c_A}{r_A}$ to be as low as possible. Thus, I would start a cult that glorifies death in war (much like the culture surrounding war in WW2 Japan) to decrease the cost of fighting, and I would pay my soldiers more! Commander $B$ would do the same with his soldiers, since he wants $\theta^*$ to be low. ■